# Original 8th Version

**P1-1.** Equation 1-1 gives a formula for the end-to-end delay of sending one packet of length $L$ over $N$ links of transmission rate $R$. Generalize this formula for sending $P$ such packets back-to-back over the $N$ links.

$$d_{end-to-end} = N\frac{L}{R} \tag{1-1}$$

**P1-2.** Consider an application that transmits data at a steady rate (for example, the sender generates an $N$-bit unit of data every $k$ time units, where $k$ is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:

a. Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?

b. Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?

**P1-3.** Consider the circuit-switched network in Figure 1.1. Recall that there are four circuits on each link. Label the four switches A, B, C, and D, going in the clockwise direction.

a. What is the maximum number of simultaneous connections that can be in progress at any one time in this network?

b. Suppose that all connections are between switches A and C. What is the maximum number of simultaneous connections that can be in progress?

c. Suppose we want to make four connections between switches A and C, and another four connections between switches B and D. Can we route these calls through the four links to accommodate all eight connections?

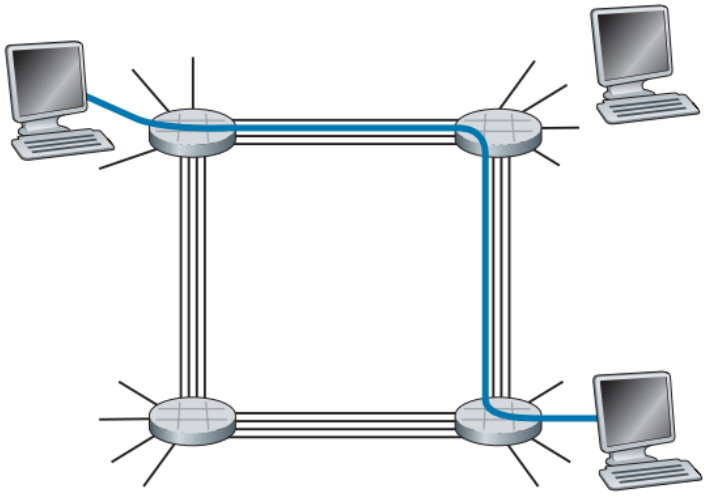


**Figure 1.1** A simple circuit-switched network consisting of four switches and four links

**P1-4.** Review the car-caravan analogy in Section 1.4. Assume a propagation speed of 100 km/hour.

a. Suppose the caravan travels 175 km, beginning in front of one tollbooth, passing through a second tollbooth, and finishing just after a third tollbooth. What is the end-to-end delay?

b. Repeat (a), now assuming that there are eight cars in the caravan instead of ten.

**P1-5.** This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate $R$ bps. Suppose that the two hosts are separated by $m$ meters, and suppose the propagation speed along the link is $s$ meters/sec. Host A is to send a packet of size $L$ bits to Host B.

a. Express the propagation delay, $d_{prop}$, in terms of $m$ and $s$.

b. Determine the transmission time of the packet, $d_{trans}$, in terms of $L$ and $R$.

c. Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.

d. Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{trans}$, where is the last bit of the packet?

e. Suppose $d_{prop}$ is greater than $d_{trans}$. At time $t = d_{trans}$, where is the first bit of the packet?

f. Suppose $d_{prop}$ is less than $d_{trans}$. At time $t = d_{trans}$, where is the first bit of the packet?

g. Suppose $s = 2.5 \times 10^8$, $L = 1500$ bytes, and $R = 10$ Mbps. Find the distance $m$ so that $d_{prop}$ equals $d_{trans}$.

**P1-6.** In this problem, we consider sending real-time voice from Host A to Host B over a packet-switched network (VoIP). Host A converts analog voice to a digital 64 kbps bit stream on the fly. Host A then groups the bits into 56-byte packets. There is one link between Hosts A and B; its transmission rate is 10 Mbps and its propagation delay is 10 msec. As soon as Host A gathers a packet, it sends it to Host B. As soon as Host B receives an entire packet, it converts the packet's bits to an analog signal. How much time elapses from the time a bit is created (from the original analog signal at Host A) until the bit is decoded (as part of the analog signal at Host B)?

**P1-7.** Consider a packet of length $L$ that begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Let $d_i$, $s_i$,

and $R_i$ denote the length, propagation speed, and the transmission rate of link $i$, for $i = 1, 2, 3$. The packet switch delays each packet by $d_{proc}$. Assuming no queuing delays, in terms of $d_i$, $s_i$, $R_i$, $(i = 1, 2, 3)$, and $L$, what is the total end-to-end delay for the packet? Suppose now the packet is $1,500$ bytes, the propagation speed on all three links is $2.5 \times 10^8$m/s, the transmission rates of all three links are $2.5$ Mbps, the packet switch processing delay is $3$ msec, the length of the first link is $5,000$ km, the length of the second link is $4,000$ km, and the length of the last link is $1,000$ km. For these values, what is the end-to-end delay?

**P1-8.** In the above problem (P1-7), suppose $R_1 = R_2 = R_3 = R$ and $d_{proc} = 0$. Further suppose that the packet switch does not store-and-forward packets but instead immediately transmits each bit it receives before waiting for the entire packet to arrive. What is the end-to-end delay?